Predictive Experiments and Observable Operators

Predictive State Spaces from Observable Experiments

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Abstract

Building on the canonical probability space representation established in *Observational Foundations of Probability*, this work studies the predictive structure that emerges from compatible observable experiments.

Given a future observable $F : \Omega \to O_F$, prediction induces an equivalence relation on observable states determined by equality of conditional laws. The resulting quotient defines a minimal predictive query Q_* whose states correspond exactly to distinct predictive laws of F. A canonical positive linear operator

$$(K_{Q_*}g)(q_*) = E[g(F) \mid Q_* = q_*]$$

then acts on functions of this predictive state space, providing an observable representation of prediction expressed entirely in terms of conditional laws. The minimal predictive query is universal: any other predictive statistic factors through Q_* via the Doob–Dynkin lemma, making Q_* the coarsest observable representation retaining the full predictive law of F.

Relation to the observational foundations. *Observational Foundations of Probability* establishes that compatible observable experiments admit a canonical probability space $(\Omega, \sigma(\mathcal{Q}), P)$. The present work studies the predictive structure that emerges once this representation is available: prediction induces an equivalence on observable states whose quotient yields a minimal predictive query Q_*. Conditional expectations of future observables therefore factor through this predictive state space.

The predictive structure developed in this note proceeds through the following sequence of constructions.

The logical progression of the present work is:

canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$

$\downarrow$

future observable $F : \Omega \to O_F$

$\downarrow$

predictive kernels $\Pi_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$

$\downarrow$

predictive equivalence $q \sim q' \iff \Pi_Q(q, \cdot) = \Pi_Q(q', \cdot)$

$\downarrow$

minimal predictive query $Q_* : \Omega \to O_{Q_*}$

$\downarrow$

predictive operator $K_{Q_*}g(q_*) = E[g(F) \mid Q_* = q_*]$

Main theorem. The principal result of this work identifies the minimal observable state space required for prediction.

Theorem 1 (Predictive state theorem)**.** *Let $F : \Omega \to O_F$ be a future observable. Predictive equivalence induces a minimal predictive query Q_* whose observable states correspond exactly to distinct predictive laws of F. In particular, conditional expectations of future observables factor through this predictive state space.*

Structure of the paper. Section 1 recalls the observational framework and the canonical probability space representation. Section 2 introduces query experiments and the experiment preorder induced by prediction. Section 3 introduces predictive kernels and the kernel-law map. Sections 4 and 5 develop predictive equivalence, the minimal predictive query, and the associated factorization theorem, followed by predictive sufficiency. Sections 6 and 7 introduce the predictive operator representation and establish operator closure on the minimal predictive query — the central theoretical result of the paper.

1 Standing framework

Assume the observational construction of *Observational Foundations of Probability.*

In particular:

- A query system $(\mathcal{Q}, \preceq)$.
- The canonical experiment
$$(\Omega, \sigma(\mathcal{Q}), P).$$
- Evaluation maps
$$Q : \Omega \to O_Q.$$
- A future observable: a measurable map
$$F : \Omega \to O_F$$
with O_F standard Borel, representing a quantity whose distribution is to be predicted from present queries.

Observable functions $g : O_F \to \mathbb{R}$ are taken to be bounded and measurable throughout. Boundedness ensures integrability of g under every predictive law (since predictive laws are probability measures) and allows conditional expectations and operator compositions to be handled by standard Bochner integration without additional integrability hypotheses.

All other results in this note are derived relative to this observational structure. The sequel develops the predictive structure that becomes available once the observational experiment (Ω, P) has been constructed.

Remark 1 (Program context)**.** *The present paper studies the predictive structure that emerges once the observational representation theorem has produced the canonical experiment* $(\Omega, \sigma(\mathcal{Q}), P)$*.*

Remark 2 (Formalization)**.** *The structures used here—query systems, refinement maps, and the canonical experiment* (Ω, P)*—correspond to the objects defined in the Lean formalization of the QuerySystem framework. In particular, evaluation maps*

$$Q : \Omega \to O_Q$$

are the measurable coordinate projections from the projective-limit space constructed in the extension theorem.

The assumption that O_F is standard Borel is used not only for the existence of the predictive kernel Π_Q (Rokhlin disintegration) but also for the measurability of the predictive law map ϕ_Q into $\mathcal{P}(O_F)$: since Π_Q is a Markov kernel, the map $q \mapsto \Pi_Q(q, \cdot)$ is measurable into $\mathcal{P}(O_F)$ equipped with the Giry σ-algebra (which on standard Borel spaces coincides with the weak convergence σ-algebra). All results involving ϕ_Q and Q_* are derived relative to this measurability.

For every future observable

$$F : \Omega \to O_F$$

regular conditional probabilities define predictive kernels

$$\Pi_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$$

for each query $Q : \Omega \to O_Q$. These kernels are compatible with refinement of queries: whenever $Q_1 \preceq Q_2$,

$$\Pi_{Q_1} = \Pi_{Q_2} \circ \pi^{Q_2}_{Q_1}.$$

The remainder of the paper develops the predictive consequences of these kernels, culminating in the minimal predictive query Q_* and its associated operator representation.

2 Query experiments and the information order

Each query induces a statistical experiment in the sense of Blackwell and Le Cam.

Definition 1 (Query experiment)**.** *For a query Q, the pushforward*

$$Q : (\Omega, P) \to (O_Q, Q_\# P)$$

defines a statistical experiment.

The refinement preorder on queries corresponds to the standard informativeness order for experiments.

Theorem 2 (Experiment preorder)**.** *If*

$$Q_1 \preceq Q_2$$

then the experiment generated by Q_1 is a measurable post-processing of the experiment generated by Q_2. In particular there exists a measurable map

$$\pi^{Q_2}_{Q_1} : O_{Q_2} \to O_{Q_1}$$

such that

$$Q_1 = \pi^{Q_2}_{Q_1} \circ Q_2.$$

Thus the query system $(\mathcal{Q}, \preceq)$ may be interpreted as a preorder of statistical experiments ordered by informativeness.

Remark 3. *The present work studies the predictive structure induced on the canonical experiment (Ω, P) by future observables.*

Remark 4. *The experiment preorder provides the structural order in which later notions of predictive sufficiency and minimality will be expressed.*

3 Predictive experiments

Definition 2 (Predictive kernel)**.** *For a query Q, the* predictive kernel *is the regular conditional distribution of F given Q:*

$$\Pi_Q(q, A) = P(F \in A \mid Q = q), \qquad q \in O_Q,\ A \in \mathcal{B}(O_F).$$

Since O_F is standard Borel, a regular conditional distribution exists, so Π_Q is a well-defined Markov kernel

$$\Pi_Q : O_Q \times \mathcal{B}(O_F) \to [0, 1].$$

The kernel determines the predictive law map

$$\phi_Q : O_Q \to \mathcal{P}(O_F), \qquad \phi_Q(q) = \Pi_Q(q, \cdot).$$

Measurability of ϕ_Q follows from the fact that Π_Q is a kernel: for each $A \in \mathcal{B}(O_F)$ the map $q \mapsto \Pi_Q(q, A)$ is measurable, so ϕ_Q is measurable into $\mathcal{P}(O_F)$ equipped with the weak convergence σ-algebra.

Remark 5 (Regular conditional distribution)**.** *The kernel Π_Q satisfies*

$$\Pi_Q(Q(\omega), A) = P(F \in A \mid Q)(\omega) \qquad P\text{-a.s.}$$

It is defined $Q_{\#}P$-almost surely in q.

Proposition 1 (Predictive compatibility)**.** *If $Q_1 \preceq Q_2$ then*

$$\Pi_{Q_1} = \Pi_{Q_2} \circ \pi^{Q_2}_{Q_1} \qquad Q_{1\#}P\text{-a.s.}$$

Proof. By the tower property of conditional expectation, for any $A \in \mathcal{B}(O_F)$,

$$P(F \in A \mid Q_1) = E\big[P(F \in A \mid Q_2) \bigm| Q_1\big] = E\big[\Pi_{Q_2}(Q_2, A) \bigm| Q_1\big].$$

Since $Q_1 = \pi^{Q_2}_{Q_1} \circ Q_2$ and $\Pi_{Q_2}(Q_2, A)$ is $\sigma(Q_2)$-measurable, the right-hand side equals $\Pi_{Q_2}(\pi^{Q_2}_{Q_1}(Q_1), A) = (\Pi_{Q_2} \circ \pi^{Q_2}_{Q_1})(Q_1, A)$. □

This expresses that predictive experiments inherit the refinement order on queries.

Remark 6 (Necessity of the a.e. qualifier)**.** *The equality $\Pi_{Q_1} = \Pi_{Q_2} \circ \pi^{Q_2}_{Q_1}$ holds $Q_{1\#}P$-almost surely, not pointwise. This is unavoidable: the regular conditional distribution $\Pi_Q(q, \cdot)$ is only determined $Q_{\#}P$-almost surely in q, since two versions of a conditional distribution may differ on a set of $Q_{\#}P$-measure zero. Any two regular conditional distributions of F given Q therefore agree $Q_{\#}P$-a.s., and the compatibility identity can only be asserted in this almost-sure sense.*

Remark 7 (Predictive statistics)**.** *From the perspective of statistical experiment theory, predictive queries behave like statistics of the present relative to the future. The minimal predictive query introduced later will therefore play the role of a dynamical sufficient statistic: the smallest observable representation of the present retaining exactly the predictive information relevant to the future observable F.*

The next step is to identify when two observable states determine the same predictive law. This leads to the notion of predictive equivalence and to the construction of the minimal predictive query.

4 Minimal predictive statistics

Definition 3 (Predictive equivalence)**.** *Let $Q : \Omega \to O_Q$ be a query. Two observable states $q, q' \in O_Q$ are said to be* predictively equivalent *if*

$$\phi_Q(q) = \phi_Q(q'),$$

or equivalently

$$\Pi_Q(q, \cdot) = \Pi_Q(q', \cdot).$$

Two observable states are predictively equivalent whenever they have the same image under ϕ_Q, i.e. whenever they induce the same predictive law of F.

Theorem 3 (Minimal predictive query)**.** *Assume O_Q and O_F are standard Borel. Define*

$$Q_* := \phi_Q \circ Q : \Omega \to \mathcal{P}(O_F).$$

Then Q_* is measurable, and

$$Q_*(\omega) = P(F \in \cdot \mid Q = Q(\omega)).$$

Two realizations satisfy $Q_*(\omega) = Q_*(\omega')$ if and only if they are predictively equivalent. The state space of Q_* is

$$O_{Q_*} = \phi_Q(O_Q) \subseteq \mathcal{P}(O_F),$$

the image of the predictive law map, so each state of Q_* corresponds exactly to a distinct predictive law of F.

Proof. Measurability of $Q_* = \phi_Q \circ Q$ follows from measurability of ϕ_Q (Definition 2) and of Q. The identification $Q_*(\omega) = P(F \in \cdot \mid Q = Q(\omega))$ is immediate from the definition of ϕ_Q. Two realizations satisfy $Q_*(\omega) = Q_*(\omega')$ iff $\phi_Q(Q(\omega)) = \phi_Q(Q(\omega'))$ iff they are predictively equivalent (Definition 3). The state space $O_{Q_*} = \phi_Q(O_Q)$ by definition. □

Remark 8 (Predictive state space)**.** *The state space $O_{Q_*} = \phi_Q(O_Q) \subseteq \mathcal{P}(O_F)$ is a space of probability measures: each state is the predictive law of F associated with a present observation. The minimal predictive query organizes prediction by mapping each observable state to its associated predictive law.*

The predictive law of F therefore depends on ω only through $Q_*(\omega)$. The following theorem makes this factorization precise and establishes its universal minimality.

Theorem 4 (Predictive factorization theorem)**.** *Let $Q_* = \phi_Q \circ Q$ be the minimal predictive query of Theorem 3, and let $\Phi : O_{Q_*} \to \mathcal{P}(O_F)$ be the identity inclusion $\Phi(q_*) = q_*$.*

Then

$$P(F \in \cdot \mid Q_* = q_*) = \Phi(q_*) = q_*,$$

so the predictive law $\Pi(\omega) = P(F \in \cdot \mid \omega)$ factors as

$$\Pi = \Phi \circ Q_*.$$

Moreover Q_* is universal: if $R : \Omega \to O_R$ is any query and $\psi : O_R \to \mathcal{P}(O_F)$ is measurable with $\Pi = \psi \circ R$, then Q_* is measurable with respect to $\sigma(R)$, and there exists a measurable map $\theta : O_R \to O_{Q_*}$ such that $Q_* = \theta \circ R$.

In particular the following diagram commutes:

$$\begin{array}{ccc} \Omega & \xrightarrow{R} & O_R \\ \downarrow_{Q_*} & & \downarrow_{\psi} \\ O_{Q_*} & \xrightarrow{\Phi} & \mathcal{P}(O_F) \end{array}$$

Thus Q_* is the minimal observable statistic through which the predictive law of the future experiment factors.

Proof. Factorization. Since $Q_*(\omega) = \phi_Q(Q(\omega)) = \Pi_Q(Q(\omega), \cdot) = P(F \in \cdot \mid Q)(\omega)$, we have $P(F \in \cdot \mid Q_* = q_*) = q_* = \Phi(q_*)$, so $\Pi = \Phi \circ Q_*$.

Universality. Suppose $\Pi = \psi \circ R$. Then $Q_* = \Pi = \psi \circ R$ is $\sigma(R)$-measurable. By the Doob–Dynkin lemma applied to the measurable map $Q_* : \Omega \to \mathcal{P}(O_F)$ and the σ-algebra $\sigma(R)$, there exists a measurable $\theta : O_R \to O_{Q_*}$ with $Q_* = \theta \circ R$.

Minimality. The universality step shows Q_* factors through any other predictive statistic. Therefore Q_* is the coarsest measurable function of ω carrying the full predictive law. □

Corollary 1 (Canonical predictive operator)**.** *For every bounded measurable $g : O_F \to \mathbb{R}$, define*

$$(Kg)(q_*) = \int g(f)\, q_*(df) = E[g(F) \mid Q_* = q_*].$$

Then $K : L^\infty(O_F) \to L^\infty(O_{Q_*})$ is a well-defined positive linear operator.

Proof. Since $q_* \in \mathcal{P}(O_F)$ is a probability measure, integration against q_* is a positive linear functional. The function $q_* \mapsto \int g\, dq_*$ is measurable in q_* by measurability of ϕ_Q and boundedness of g. Positivity and linearity of K are inherited from those of integration. □

Remark 9 (Predictive operators)**.** *The predictive factorization therefore induces operators acting on functions of the predictive state. These operators represent the evolution of predictive expectations of future observables.*

In particular, if F_t denotes the observable at prediction horizon t, the operators

$$(K_t g)(q_*) = E[g(F_t) \mid Q_* = q_*]$$

form the predictive evolution operators studied in the next paper.

Remark 10 (Predictive horizon quotient)**.** *The canonical experiment space Ω may be interpreted as an observable horizon of histories. Predictive equivalence identifies two horizon configurations whenever they induce the same conditional law of the future observable. The minimal predictive query Q_* is therefore the quotient of the observable horizon by predictive indistinguishability. In this sense Q_* is the minimal observable boundary state needed for autonomous prediction.*

Remark 11 (Lattice interpretation)**.** *When the experiment preorder admits infima, Q_* equals the meet of all predictively sufficient queries:*

$$Q_* = \bigwedge_{Q \in \mathcal{S}} Q.$$

Thus Q_* is the largest common coarsening of all sufficient experiments: it extracts exactly the predictive information shared by all of them, no more.

Remark 12 (Relation to Blackwell sufficiency)**.** *In the language of statistical experiment theory, the minimal predictive query plays the role of the minimal sufficient experiment for predicting the future observable F: it is the coarsest observable statistic that retains the full predictive law.*

Remark 13 (Doob–Dynkin factorization of the future experiment)**.** *The predictive factorization theorem (Theorem 4) shows that the minimal predictive query Q_* is the minimal Doob–Dynkin factor through which the future experiment factors. In this sense Q_* is simultaneously a minimal sufficient statistic and a minimal Doob–Dynkin factor for prediction.*

5 Predictive sufficiency

The minimal predictive query constructed in the previous section captures all predictive information about the future observable. This section interprets that construction through the notion of predictive sufficiency and relates it to classical ideas from statistical experiment theory.

Definition 4 (Predictive sufficiency). *A query Q is predictively sufficient if for every admissible query Q' there exists a measurable map*

$$\psi : O_Q \to O_{Q'}$$

such that

$$\Pi_{Q'} = \Pi_Q \circ \psi.$$

Predictively sufficient queries therefore retain all predictive information about the future observable.

Theorem 5 (Predictive sufficiency characterization). *A query Q is predictively sufficient if and only if*

$$F \perp Q' \mid Q$$

for all admissible queries Q'.

Proof. ($\Rightarrow$) Suppose Q is predictively sufficient, so for every Q' there exists measurable $\psi : O_Q \to O_{Q'}$ with $\Pi_{Q'} = \Pi_Q \circ \psi$. Then for any $A \in \mathcal{B}(O_F)$,

$$P(F \in A \mid Q, Q') = P(F \in A \mid Q),$$

which is exactly $F \perp Q' \mid Q$.

($\Leftarrow$) If $F \perp Q' \mid Q$ for all Q', then for any A, $P(F \in A \mid Q, Q') = P(F \in A \mid Q)$. In particular $P(F \in A \mid Q') = E[P(F \in A \mid Q) \mid Q'] = E[\Pi_Q(Q, A) \mid Q']$, so $\Pi_{Q'} = \Pi_Q \circ \psi$ for the conditional expectation map $\psi(q') = E[Q \mid Q' = q']$ (when Q is a function of Q' this is the refinement map; in general it is a regular conditional kernel). Thus Q is predictively sufficient. ☐

Remark 14 (Operator formulation and boundedness). *In the operator formulation of Section 6, predictive sufficiency of Q implies that for every bounded measurable $g : O_F \to \mathbb{R}$,*

$$E[g(F) \mid Q] = E[g(F) \mid Q_*] \qquad P\text{-a.s.}$$

The Lean formalization of this statement (`predictive_sufficiency`) requires an explicit uniform bound on g for the Bochner integration step.

6 Predictive operators

Definition 5 (Predictive operator). *For a query Q and bounded measurable function $g : O_F \to \mathbb{R}$, define*

$$(K_Q g)(q) = E[g(F) \mid Q = q].$$

Proposition 2 (Predictive operator well-definedness). *For bounded measurable g, K_Q defines a positive linear operator.*

Theorem 6 (Predictive operator representation). *Let $F : \Omega \to O_F$ be a future observable and let Q_* be the minimal predictive query. Let $g : O_F \to \mathbb{R}$ be a bounded measurable function. Then the map*

$$(K_{Q_*} g)(q_*) = E[g(F) \mid Q_* = q_*]$$

defines a Markov operator on the predictive state space.

Moreover, the predictive evolution of observable expectations factors through Q_* and is therefore represented entirely by the operator K_{Q_*}.

Remark 15 (Boundedness in the Lean formalization)**.** *The Lean formalization of this theorem (`predictive_factorization` and `predictive_sufficiency`) carries an explicit uniform bound $\exists C, \forall f, |g(f)| \le C$ as a hypothesis. This is needed because the proof uses Bochner integration against a family of probability measures indexed by q_*, and boundedness ensures integrability under every predictive law without further verification. Boundedness of g is listed as a standing assumption in Section 1, so the hypothesis is always satisfied in practice.*

Remark 16 (Kernel representation)**.** *The predictive operator is the integral operator induced by the predictive kernel:*

$$(K_Q g)(q) = \int g(f)\, \Pi_Q(q, df).$$

Thus predictive operators form a class of Markov operators on the observable state space. In particular, when $Q = Q_*$, the predictive operator K_{Q_*} gives the canonical operator representation of the future experiment on the minimal predictive state space.

Proposition 3 (Koopman deterministic specialization)**.** *If the predictive kernel is deterministic,*

$$\Pi_Q(q) = \delta_{\phi(q)},$$

then

$$(K_Q g)(q) = g(\phi(q)).$$

Thus classical Koopman operators arise as a special case.

7 Operator closure

Corollary 2 (Observable operator closure)**.** *If Q_* is predictively sufficient then*

$$K_{Q_*} g(q) = E[g(F) \mid Q_* = q]$$

completely determines the predictive evolution of observable expectations. In particular, predictive dynamics may be represented entirely as an operator acting on the predictive state space Q_*.

Remark 17 (Representation of predictive dynamics)**.** *The minimal predictive query Q_* provides an observable state space for prediction: its states correspond exactly to distinct conditional laws of the future observable F. The predictive operator K_{Q_*} therefore describes the evolution of predictive expectations on this state space, yielding an observable representation of predictive dynamics without reference to latent state variables.*

8 Conceptual chain

The preceding sections establish the following structural progression.

query system

$\downarrow$

probabilistic representation

$\downarrow$

query experiments and experiment preorder

$\downarrow$

predictive experiments and predictive kernels

$\downarrow$

predictive equivalence

$\downarrow$

Q_* (minimal predictive state space)

$\downarrow$

K_{Q_*} (predictive operator on O_{Q_*})

Remark 18 (Continuation)**.** *The horizon-indexed family $\{K_t\}_{t\geq 0}$, its semigroup structure, the infinitesimal generator, and Koopman–Perron duality are developed in* Predictive Operator Theory for Observable Dynamical Systems*. The constructive route to the predictive state via delay embeddings — including the probabilistic sufficiency criterion and the Takens specialization — is developed in* Observational Probability from Delay Queries*.*

9 Conclusion and related work

What has been established. Starting from the canonical probability space $(\Omega, \sigma(\mathcal{Q}), P)$ of *Observational Foundations of Probability*, this paper has developed the predictive structure that emerges from a future observable $F : \Omega \to O_F$. The main results are:

1. *Predictive kernels.* Each query Q induces a conditional kernel $\Pi_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$, giving prediction a concrete query-level representation.

2. *Minimal predictive query.* Predictive equivalence $q \sim q'$ iff $\Pi_Q(q, \cdot) = \Pi_Q(q', \cdot)$ factors Q through a minimal statistic Q_* whose states are in bijection with distinct predictive laws of F.

3. *Doob–Dynkin universality.* Any measurable predictive statistic factors through Q_*, making the minimal predictive query the coarsest observable representation retaining full predictive content.

4. *Predictive operator.* Conditional expectation $(K_{Q_*}g)(q_*) = E[g(F) \mid Q_* = q_*]$ defines a positive linear contraction on O_{Q_*} that is closed under the predictive state algebra.

Relation to classical sufficiency. The minimal predictive query Q_* is an instance of a sufficient statistic in the sense of Fisher and Halmos–Savage: it retains all information relevant to predicting F while discarding information that is irrelevant. The framework of Blackwell [1] and Le Cam [3] on comparison and sufficiency of statistical experiments provides the conceptual background; the present work specialises their comparison order to the query setting and identifies the minimal representative in a canonical way via the Doob–Dynkin lemma. The experiment preorder induced by prediction is a structural refinement of the Blackwell order restricted to the observable interface.

Relation to de Finetti and predictive sufficiency. In de Finetti's representation theorem [2], the exchangeable measure is disintegrated over an invariant σ-field that plays the role of the predictive state. The present framework provides a direct construction of that σ-field from the observable query structure, without requiring an exchangeability or invariance assumption. Predictive sufficiency (Section 5) is the query-theoretic counterpart of the Radon–Nikodym disintegration underlying de Finetti's theorem.

What follows. The predictive operator K_{Q_*} established here is a single-horizon object. *Predictive Operator Theory for Observable Dynamical Systems* (Paper 3) promotes this to a horizon-indexed family $\{K_t\}_{t\geq 0}$ and shows it forms a strongly continuous Markov operator semigroup, from which an infinitesimal generator and predictive Kolmogorov equation follow. The constructive route to Q_* via delay embeddings of a dynamical system — and the connection to Takens' embedding theorem — is the subject of *Observational Probability from Delay Queries* (Paper 4).

References

[1] David Blackwell. Comparison of experiments. In *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, pages 93–102, 1951.

[2] Bruno de Finetti. *Theory of Probability.* Wiley, 1974.

[3] Lucien Le Cam. *Asymptotic Methods in Statistical Decision Theory.* Springer, 1986.

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